8-Queen Problem

AI-2.py - C:/Users/Vineela/OneDrive/Documents/AI-2.py (3.15.0)

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```
N = 8
board = [-1] * N

def solve(row):
    if row == N:
        return True

    for col in range(N):
        safe = True

        for r in range(row):
            if board[r] == col or abs(board[r] - col) == abs(row - r):
                safe = False
                break

        if safe:
            board[row] = col

            if solve(row + 1):
                return True

            board[row] = -1

    return False

if solve(0):
    print("8-Queens Solution:")
    for row in range(N):
        print(" ".join("Q" if board[row] == col else "-"
                       for col in range(N)))
else:
    print("No solution found.")
```

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```
Python 3.15.0 (tags/v3.15.0:d220409, Oct  9 2026, 11:55:56) [MSC v.1951 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Vineela/AppData/Local/Programs/Python/Python315/AI-1.py ==
Solution found:
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

>>>
================= RESTART: C:/Users/Vineela/OneDrive/Documents/AI-2.py =================
8-Queens Solution:
Q - - - - - - -
- - - - Q - - -
- - - - - - - Q
- - - - - Q - -
- - Q - - - - -
- - - - - - Q -
- Q - - - - - -
- - - Q - - - -
>>>
```